Module 4

Short Answers

1. Define traction in an electric vehicle.
2. What is meant by tractive effort?
3. List the forces opposing the motion of an electric vehicle.
4. Define rolling resistance.
5. State any three factors affecting rolling resistance.
6. What is aerodynamic resistance?
7. Why does aerodynamic drag increase at higher speeds?
8. Define gradient resistance.
9. Write the expression for acceleration force.
10. Write the equation for total tractive effort.
11. What is the significance of the tractive effort-speed characteristic?
12. Define Traction Control System (TCS).
13. State the condition for vehicle acceleration.
14. State the condition for constant-speed operation.
15. Define electric vehicle driving range.
16. Write the equation used for EV range estimation.
17. Define tractive power and write its expression.
18. What is transmission efficiency in an electric vehicle?
19. What is a skateboard chassis?
20. List any four challenges in electric vehicle design.

Long Answers

1. Explain traction and tractive effort in electric vehicles. Discuss the various forces opposing vehicle motion with suitable equations.
2. Derive the mathematical model of vehicle acceleration considering rolling resistance, aerodynamic resistance, gradient resistance, and acceleration force.
3. Explain the tractive effort-speed characteristics of an electric vehicle with a neat sketch and discuss the maximum speed condition.
4. Describe the working principle, functions, and advantages of a Traction Control System (TCS) in electric vehicles.
5. Derive the expressions for rolling resistance, aerodynamic resistance, gradient resistance, and acceleration force. Explain their effects on EV performance.

6. Explain electric vehicle range modelling and derive the equation for estimating the driving range of an EV.
7. Discuss tractive power, efficiency, and energy consumption in electric vehicles with relevant equations and examples.
8. Explain aerodynamic considerations in electric vehicles. Discuss drag force, aerodynamic power requirement, and factors affecting aerodynamics.
9. Describe various aerodynamic design features used in modern electric vehicles to improve efficiency and driving range.
10. Explain transmission efficiency in electric vehicles. Discuss drivetrain efficiency, energy flow, losses, and typical efficiency values.
11. Explain the objectives of electric vehicle chassis design and discuss the different types of chassis used in EVs with their advantages.
12. Discuss the design considerations of electric vehicle body structures, including lightweight materials, battery integration, thermal management, safety, and aerodynamics.
13. Explain the major issues and challenges in electric vehicle design such as battery-related issues, thermal management, weight management, charging infrastructure, safety concerns, NVH, and cost constraints.